

Raise the Roof?

The King County Housing Authority adds insulation and slope while removing ventilation to make a more energy-efficient and durable roof.

by Dan Auer and Claude Iosso

Across the Northwest, flat roofs top apartment buildings, office complexes, and supermarkets. Cheap and easy to build, flat roofs are the obvious option for boxlike commercial and multifamily buildings. Unfortunately, they can be a bad choice in a cool, wet climate. The problem can be acute in the Northwest, particularly west of the Cascade Mountains, where it usually rains and stays cool from October to July. All that water has a tendency to pool on flat roofs and finds ways to penetrate any roof surface, whether it's tar, bitumen, rubber, asphalt, or metal.

In addition to the risks from rainwater leaking in, flat roofs in the Northwest are susceptible to risks from other sources of moisture as well. Mandated by most city building codes, attic ventilation can be effective in some climates. Breather vents designed to draw condensation from the attic space to the exterior can do just the opposite in the Northwest, due to high outdoor relative humidity (RH). Humid air and moisture can accumulate in the attic cavity. Condensation occurs when the warm air from a building's interior meets the relatively cold sheathing and joists under the roof membrane.

As if susceptibility to harmful moisture weren't enough of a problem, ventilated, lightly insulated flat roofs do a poor job of keeping heat in the buildings they cover. The roofs are typically framed with the rafter doubling as the joist. The typical flat roof has less than 1 ft of attic space, allowing for only a few inches of insulation—usually 1 or 2 inches of fiberglass batt that may have been exposed to extreme temperatures and high RH for 30 years. Even if a building has well-sealed double-pane windows, if heat is pouring out of the roof, property owners or renters are losing money in utility costs.



Rigid insulation is installed as part of a new roof.

The Seattle Public Utilities Sustainable Building Division says that flat and low-slope roofs are “notoriously leaky, and replacing them usually generates a tremendous amount of waste. They’re also costly to repair.” To put it flatly, flat roofs and rainy climates do not mix.

Case in Point at Bellevue

With an inventory of more than 100 apartment complexes, about half of them with flat roofs, our employer, the King County Housing Authority (KCHA), is justifiably concerned about flat roofs’ durability and thermal performance. These roofs let heat from the building escape into the gray sky. When the roofs at a KCHA complex in Bellevue, Washington, kept failing prematurely, KCHA began to consider ways to modify their design.

All of the typical problems with flat roofs in the Northwest have plagued Cascadian Apartments, a 198-unit com-

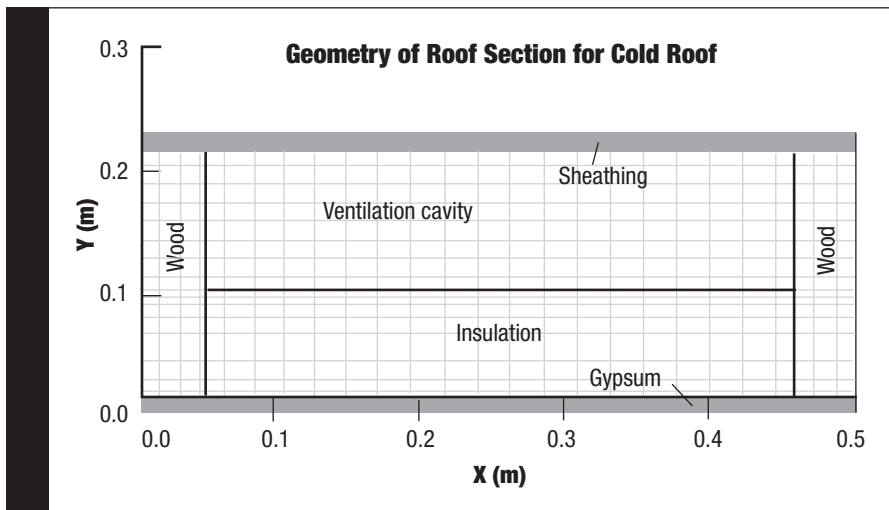
plex in Bellevue. Built in 1968, Cascadian is a handsome, well-maintained property with swimming pools and attractive landscaping. There are 14 buildings at Cascadian, nearly all with traditional flat roofs. (Building H got a pitched roof replacement in 2000, but it was very expensive—\$16/ft².) The flat roofs have a simple, inexpensive design: a waterproof membrane, made of several layers of rolled composition material coated with asphalt, topping a roof deck of plywood sheathing and wood joists. The attic cavity between the ceilings of the top-floor apartments and the roof membrane is 8 inches deep. With continuous soffit venting to the outside, this space features 3 inches of R-11 batt insulation on top of the ceiling Sheetrock, leaving a 5-inch gap between the insulation and the sheathing (see Figure 1).

Rainwater pools on the roofs, finding ways over time to penetrate the asphalt membrane and leak into the roof deck. The Housing Authority inspects the



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The new insulated roof system employs tapered insulation to move rainwater off the roof.



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Figure 1. In the preexisting roof configuration for building K, Cascadian Apartments, passive ventilation through the ventilation cavity was provided through a continuous soffit at the building perimeter.

roofs regularly and repairs leaks quickly, but other sources of moisture are difficult to eliminate. Those vents designed to draw moisture away from the attic spaces at Cascadian have done the opposite. In many cases, condensation has accumulated in the sheathing, when the relatively warm air from the units meets cold surfaces.

Not only are the roofs at Cascadian susceptible to moisture problems, but they are energy inefficient. The apartments are electrically heated with radiant ceiling panels, so they have especially warm ceilings. As a result, the ventilated attics over the top-floor units act

like open windows next to the heat plant of the building.

In 2002, when moisture problems at Cascadian took their inevitable toll and the roofs on many of the complex's buildings were beginning to fail, the Housing Authority was compelled to change things. Roof repair or replacement for these buildings costs hundreds of thousands of dollars. And while KCHA does not have to absorb high utility bills directly, it is committed to green building and energy conservation. We were troubled that residents' heating bills were higher than they should be.

The Experiment at Building K

The Housing Authority could speculate as to why the roofs kept rotting out at Cascadian, but without being certain about causes, it couldn't progress to solutions. So in the summer of 2002, the KCHA's Repair and Weatherization Department consulted a team of building design experts. Andre Desjarlais is a building scientist from Oak Ridge National Laboratory (ORNL), and he is an expert on low-slope roofs. John Straube is a civil engineering professor at the University of Waterloo in Ontario who also provides consultant services for building design improvements through a firm called Balanced Solutions. A third consultant, Achilles Karagiozis, is one of Desjarlais' colleagues at ORNL. Karagiozis is a building science expert in the study of moisture migration through buildings.

According to Karagiozis, "Roofs are one of the simplest envelope systems we have in a building," but typically designers do not apply basic moisture-engineering principles during the design stage, and consequently most roofs are not durable. While that might have been acceptable a few years ago, now—with our better understanding of moisture transport and with the available software tools such as WUFI and MOISTURE-EXPERT—it has become simply unacceptable.

Straube supplied and installed sensors that track temperature, moisture, and RH throughout the roof deck of building K: at the top and bottom of the joists, in the sheathing, on top of the roof membrane, on top of the insulation, and on the top and bottom of the gypsum ceiling. Karagiozis performed advanced hygrothermal simulations to pinpoint the exact locations for sensor placement. If placed only a few millimeters from the critical regions, the sensors could provide misleading moisture readings. In addition, sensors were posted outside, to track the weather. The sensors have provided continuous readings for one year and have provided valuable data to validate the hygrothermal model (see "Conceptual Proof").

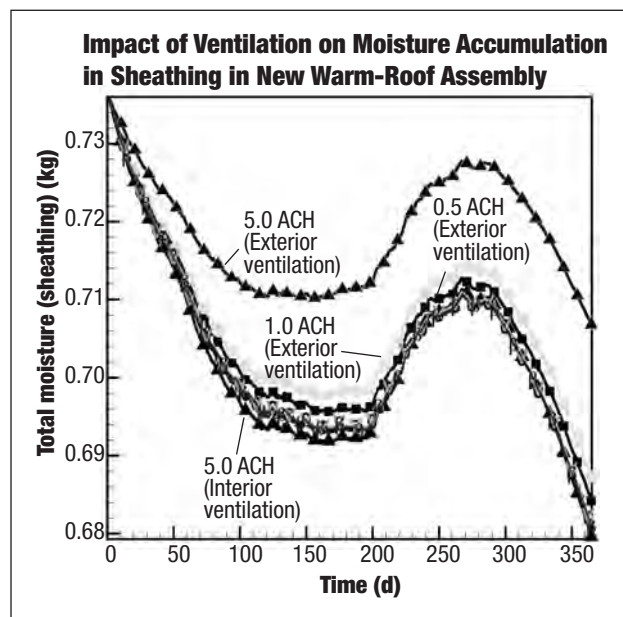
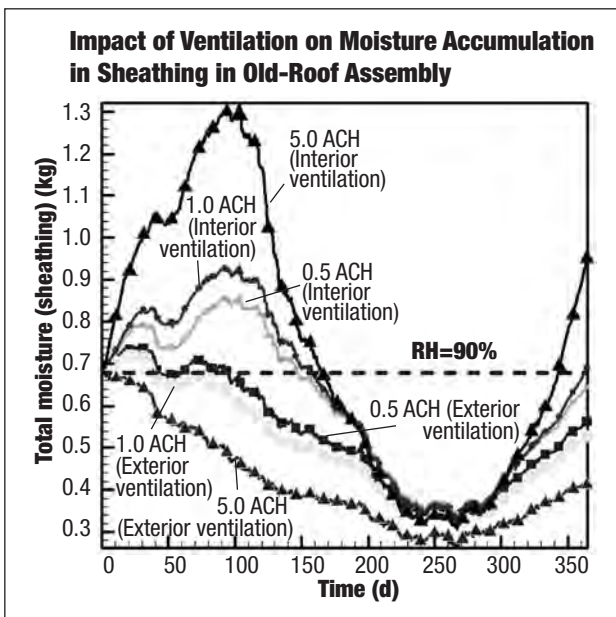
If there were any doubt that the traditional flat-roof design invited problems, a

Conceptual Proof

During the first phase of the project, computer simulation work performed using the state-of-the-art hygrothermal model MOISTURE-EXPERT by Karagiozis clearly demonstrated that moisture accumulation in the roof cassette (the sheathing board and the wood joist) is sensitive to small amounts of air leakage from the interior of the building,

and to a lack of effective drying from active ventilation (see Figure A). There is a corresponding lack of moisture sensitivity for the warm-roof design (see Figure B). For the cold roof, air leakage from the interior is the most critical. Day 1 coincides with October 1, and there is a continuous increase in moisture accumulation during

the fall and winter months. The less airtight the system is, the more moisture accumulates. The simulations predicted that peak moisture content in the sheathing board would be 42%. That is significantly higher than the maximum recommended fiber saturation of 16%.



(left) **Figure A.** The old, cold roof design allowed moisture to accumulate in the sheathing board and wood joists. (right) **Figure B.** The new, warm roof design keeps the sheathing board and wood joists dry.

year of readings from the sensors would have dispelled it. The moisture content in a roof deck should almost never reach 25%. The readings indicated that the moisture content in the sheathing and joists hovered at around 30% from November to April. According to the consultants, the only way to keep moisture from building up in flat roofs like Cascadian's would be to install an active ventilation system—a fan that would draw air from the attic cavity to the outside. A system like this would be expensive to install and would take energy to run, and it would pump the building's heat straight out. It could work only if there were an airtight plane at the ceiling interface, and if there were no stagnant regions inside the roof cavity. Simulation studies performed for KCHA demonstrated that this kind of system could be made to work, but that the cost

of running it would be prohibitive—30%–80% more expensive than another approach proposed by our consultants.

The consultants proposed that KCHA take the warm-roof approach—that is, that the new roof be built with insulation and without venting (see Figure 2). They hypothesized that sloped rigid insulation on top of the roof deck would reduce RH and moisture buildup in the attic dramatically. They believed that active ventilation would not be necessary to keep the attic dry.

Key to warm-roof design is the temperature of the sheathing. The frontier between the cold outside and the warm inside in a cool-roof system is an obvious spot for moisture to accumulate. In warm-roof design, the sheathing stays warm under a layer of rigid foam insulation. Since temperatures under the insulation remain above the dew point,

there is no condensation on the inside surface of the sheathing. Condensation cannot accumulate in the insulation above the sheathing either, because there is no air circulation there.

With no vents, however, the new design for the new roofs did not meet the city of Bellevue's building code. KCHA negotiated with city officials, explained the experiment, and received a variance. The city wanted to make sure that measures were in place to control moisture buildup. They approved a design that promised to do just that.

The rotted joists and sheathing in building K were replaced, and the vents for the attic cavity were closed up, sealing the space. Rigid insulation, averaging R-38 with a 1 to 4 slope, was installed, tapering toward the eaves, over the ceiling joists and sheathing. The tapering was expected to reduce ponding and



Moisture and temperature sensors were installed at four different locations in the attic space.

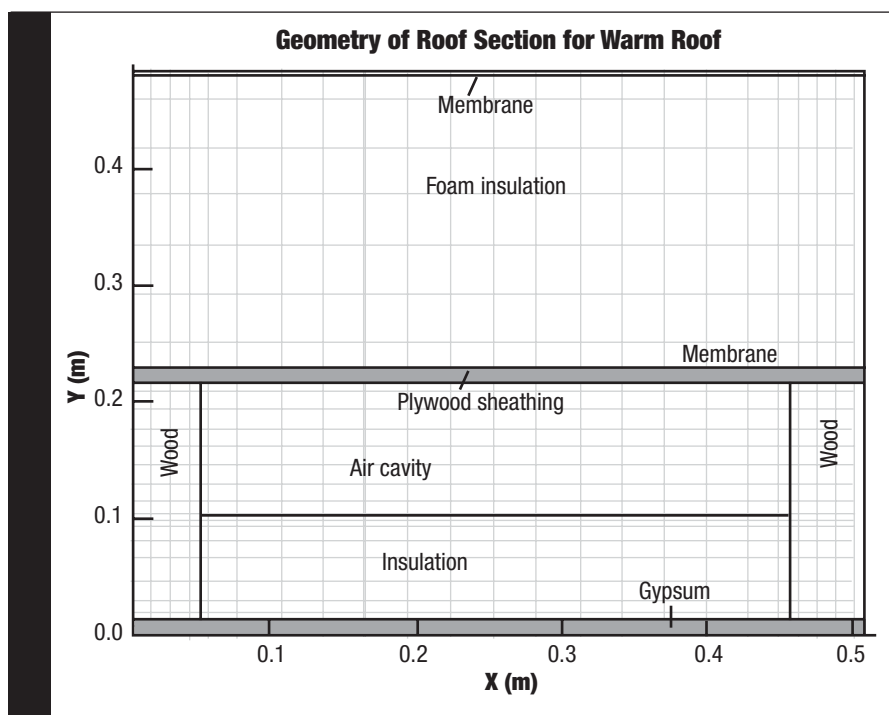


Figure 2. In the new roof configuration for building K, the attic cavity has been air sealed and insulated.

keep water out, while the insulation would keep the heat in. Moisture, temperature, and humidity sensors were reinstalled throughout the new roof deck.

Numbers Aren't a Wet Blanket

The numbers, after a year, are indeed encouraging. Whereas moisture in the

joists and sheathing of the old roof had hovered at around 30% throughout the winter months, it stayed around 16% in the warm roof. RH, which was often close to 100% in the attic cavity under the old roof, never rose much above 70% under the warm roof. The attic space essentially stayed dry. Thermal performance at building K also

improved; it used 15 %–20% less energy for heating last year than similar buildings at Cascadian that had flat roofs.

“The data are very convincing,” said Fred Campbell, director of Repair and Weatherization. “The cities of Bellevue and Federal Way are interested, and it looks like building organizations in the region and nationwide could see the wisdom of this approach.”

KCHA will incorporate the new design as it replaces more roofs at Cascadian this year. It has built warm roofs on two other large Bellevue developments, as well as on an apartment complex in the unincorporated King County neighborhood of White Center. The Repair and Weatherization Department makes improvements to private low-income housing, and it has discussed code allowances with the city of Federal Way for an apartment building there.

KCHA will continue to introduce the warm-roof design in more of its properties. In the process, it is working with some jurisdictions for code changes to allow, or even encourage, an improved design. If architects and builders warm to the idea, the way flat roofs are designed throughout the Northwest could change for the better. The potential is great for the warm-roof design in the Northwest. Besides being energy efficient, roofs built this way will last longer than traditional flat roofs.

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For more information:

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